

Sumit Raj

CONTACT

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Portfolio: sumit966-github-io.vercel.app

SKILLS

Languages: Python, SQL
LLMs and GenAI: LLMs, RAG, LangChain, LangGraph, Prompt Engineering, OpenAI API, Hugging Face, Transformers, LoRA Fine-tuning, Embeddings
Retrieval: ChromaDB, FAISS, Sentence Transformers, Hybrid Search (BM25 + Dense)
Deep Learning: PyTorch, TensorFlow, Scikit-learn
Backend and Deployment: FastAPI, Docker, GitHub Actions CI/CD, Git

EDUCATION

Visvesvaraya National Institute of Technology (VNIT), Nagpur 2024 – May 2026
M.Tech, Applied AI and Machine Learning
Dr. D.Y. Patil Institute of Technology, Pune 2019 – 2023
B.E., Computer Science CGPA: 7.89/10
Certifications: DeepLearning.AI: Generative AI with LLMs (in progress); Coursera: Deep Learning Specialization (in progress)

EXPERIENCE

Software Engineer Intern, Salesforce (Remote) Feb 2022 – May 2022

- Built Python bulk data-processing pipelines, cutting manual data handling by approximately 40%.
- Integrated Salesforce APIs into optimized backend workflows, improving system efficiency by approximately 25%.

PROJECTS

RAG-Based Clinical Question Answering System 2025
Python, LangChain, ChromaDB, FastAPI, Docker, GitHub Actions

- Built a RAG pipeline for clinical document retrieval with LangChain and ChromaDB, reaching 92% answer relevance on a 100-question benchmark.
- Added hybrid search (BM25 + dense embeddings) with Reciprocal Rank Fusion, improving retrieval accuracy by 18% over the baseline.
- Deployed with FastAPI, Docker and GitHub Actions CI/CD at under 500 ms inference latency.

LLM-Powered Security Log Analyzer 2025
Python, LangGraph, OpenAI API

- Built a LangGraph agent workflow with the OpenAI API to analyze 50K+ security logs and generate natural-language threat summaries.
- Added a self-reflection evaluation loop that kept hallucination under 3% on 200 curated test cases.

Brain Tumor Detection from MRI Scans 2024
Python, Deep Learning

- Trained a deep learning model to classify brain tumors on 3,000+ MRI scans.

Military Vehicle Detection and Face Authentication Security System Academic Project
Python, YOLOv8, OpenCV, Haar Cascade

- Built a real-time surveillance system with Haar Cascade face authentication at approximately 95% detection accuracy.
- Detected and tracked military vehicles with YOLOv8 and OpenCV, sending automated email alerts for security events.

Mobile Botnet Detection System 2024
Python, Scikit-learn

- Built a machine learning model to detect malicious bot communications with C&C servers by analyzing Android network traffic patterns.